

M3.11 Team Deliverable 2: Needfinding Report

Smart Factory Team

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Instructions:

To receive credit for your progress, please attach the following documentation.

For this module's deliverable, your team will generate a short report on needfinding activities related to your project. To do this, you must use one or more techniques we learned in Module 2 (e.g., semi-structured interviews, surveys, or questionnaires) and use them to create a list of goals, needs, and desired features—after collecting this data from people who are NOT on your team. For this deliverable, you will be focusing on the data collected and your team's interpretation of those into rough goals. Later (not in this deliverable) you will use this data to develop user stories.

Please structure your report as follows:

1. Quick summary of project—This should be borrowed and likely iterated on from your Module 2 assignment.
2. Selected needfinding method(s) and reasoning—This should describe which method you have used (e.g., semi-structured interviews, surveys, or any outside reading of Module 2) and who your “potential customers” or participants were. Do not use their names, just their profession or why they may be relevant as a customer.
3. Data collection raw results—This should be whatever material you generate. For example, transcripts of interviews, notes that summarize discussion, survey entries, etc. The objective here is to use this material to highlight important aspects of customer goals, needs, and practices.
4. Data collection highlights—This should include what your team interprets as the most important and relevant data about user goals. If possible, it would be best to summarize recurring themes and/or contrast different customer responses.
5. Initial understanding of user goals—This should enumerate the goals of your potential customers based on your interpretation of needfinding data.

Github Repository: <https://github.com/SmartFactory-Team>

Product Description:

We envision a cloud-based smart factory application that allows users to model, simulate, and optimize factory operations. For our MVP, we will focus on modeling an aluminum coil plant. We prioritize robust security features, production parameter optimization, and simulation/reporting functionality. We plan to build the UI/UX with React/JS and the backend with Python and Postgres/MySQL databases.

Needfinding Methods and Reasoning:

Method: Survey

Participant 1: Controls Engineer

Relevance: Works in a manufacturing environment so will have insight on whether or not the simulation software would be useful.

Method: Survey

Participant 2: Connected Factory Project Manager

Relevance: Manages projects focused on digitizing manufacturing environments with hardware and software.

Method: Survey

Participant 3: Random IT Student

Relevance: Worked with data analyst projects in class, have surface level understanding of factory production line

The surveys provided to the participants were slightly different to capture their unique perspectives.

Data Collection - Raw Results:

Participant 1

1. **Have you done any data analysis for a production system?** - Yes
 - a. **If yes, what kind of data analysis have you done?** - *I have done several types of data analysis including downtime event tracking, parameter definition for processes involving different products, and general root cause analysis.*
2. **Have you needed to estimate an outcome of a manufacturing process before?** - Yes, *I need to do that fairly often to see if a new product can run or to see what could be going wrong with a product.*

3. **Would it be useful to have data showing the status of the process at different steps?** - Yes
 - a. **If yes, what data points would be good to know?** - *The varying set points in the process like what the entry and incoming thickness for each step and other variables.*
4. **What are the biggest causes of downtime for the process you work on?** - *Biggest causes of downtime would be tension or thickness issues that result in a coil needing to be scrapped and removed from the mill manually.*

Participant 2

1. **What are typical challenges in industrial software implementation?** - *I think UI/UX is extremely important. Especially when tracking many different parts or steps in a process and visualizing the flow, it is important that the application remains responsive with an intuitive persona-based layout.*
2. **What are the expected benefits of industrial simulation software?** - *Simulating different process parameters and line setups should allow engineers to more easily optimize their lines without expensive experiments. Running reports based on simulated or collected data should provide an overview of operations to enable long-term planning and predictive maintenance (minimizing cost). Altogether, you can expect higher throughput, less downtime, and improved maintenance/quality control.*
3. **What are the expected risks?** - *Applying software to an industrial environment, in my experience, poses data security risks. Models/data can reveal proprietary details about operations or product specifications.*

Participant 3

1. **What aspects of a technical application's user interface help you understand complex data or system behavior?** - *I usually prefer dashboards with clear labels, color-coding, and collapsible sections. If the system deals with a lot of parameters, I like seeing step-by-step workflows.*
2. **What features would make a simulation or modeling tool easier for someone without deep manufacturing experience to use?** - *A guided setup wizard would be helpful. Something that walks the user through choosing parameters, entering data, and running the simulation. Also, tooltips that explain technical terms would help beginners understand the process.*
3. **How important is application performance and responsiveness when working with large datasets or complex simulations?** - *It's pretty important. If I click something and*

the app freezes or takes too long, I assume something is wrong. Even if the simulation takes time, the UI should stay responsive and show progress indicators.

Data Collection - Highlights:

The controls engineer we interviewed appears to already perform data analysis in his day-to-day life. He appears to calculate outcomes of process changes while also conducting frequent root cause analyses. We forecast that our software can help enhance these analysis tasks that the engineer already does. For example, we foresee the software (1) enabling improved outcomes more quickly and efficiently and (2) enabling analytics for production systems/changes that are too complex to understand with traditional tools.

One notable observation from our needfinding activities is that potential users consistently emphasized usability and clarity over advanced optimization features. This suggests that our MVP should prioritize a clear, responsive dashboard and transparent simulation outputs before expanding into more advanced capabilities.

The project manager seems to value non-functional features like UI/UX and data security while also appreciating the functional benefits of simulation and data analysis capabilities.

In accordance with the three key product attributes, the IT student highly values usability of a product. A responsive application that is consistently available is necessary for a final product. Similarly, ease of use is prioritized, lowering the technical threshold necessary to maneuver an application.

User Goals:

In our M1 report, we presented the following epics:

As a production analyst, I'd like the ability to model and optimize an aluminum production process, so I can run a line that maximizes profit.

As a production analyst, I'd like the ability to run simulations and see reports to understand the dynamics of our aluminum production process.

As an IT engineer, I'd like the ability to easily view, modify, and maintain the code for this application, so I can keep my end-users satisfied.

As a security analyst, I'd like the application to be compliant with at least 1 cybersecurity framework, so I can mitigate cyber risk.

As a production controller, I'd like the ability to connect the cloud-based production model to real-time, real-world data feeds, so I can have a digital twin that supports my decision-making.

Based on the needfinding analysis conducted in this report, we add the following user stories:

As a controls engineer, I'd like the ability to set parameter values for different stages of the process, so my simulations translate into real decisions.

As a controls engineer, I'd like to see the outcome of a simulation in clear terms, so I can more accurately estimate the impact of a change on the process's performance.

As a controls engineer, I'd like to see control charts corresponding with a simulated operation, so I can better estimate the impact of a change on the frequency of defects like coil thickness and tension issues.

As a project manager, I'd like the application to have a configurable, intuitive user interface, so end users do not have issues adopting it in their daily work.

As a project manager, I'd like the application to perform well under high data load, so end users are not slowed down by the application's performance.

Scrum Reports:

Meeting time: February 12th, 2026

Present for the Meeting (All Team Members)

- Hoang Nguyen
- William Miller
- Chase Post
- Shiloh Parker
- Thierry Tran

Completed Actions

- Finished the Waterfall Plan Team Deliverable.
- Decided next meeting date.

Decisions / Assigned Tasks

- We decided as a group that we prefer being in Section 1 (2/19 start date)
- Individually brainstorm survey questions and align over the weekend.
- Finish and populate needfinding report/survey by Thursday 2/19, when sprint 1 begins.

Summary / TO-DO

- Create a series of questions to gather information for a Needfinding Report (preferably 6-7 questions)